Today: Sparse regression/compressive sensing.

Setup $Y = X\beta^*$, $\beta^* \in \mathbb{R}^p$ (p = # of features).
$X \in \mathbb{R}^{n \times p}$. (Noiseless setting).

Goal Recover β^* from (Y, X).

Sparsity $\|\beta^*\|_0 = s$, $\|\beta^*\|_0 = |i : \beta_i^* \neq 0|$.

High-Dimensions: $n \ll p$, $n, p \to \infty$..

n equations
p unknown, $n \ll p$

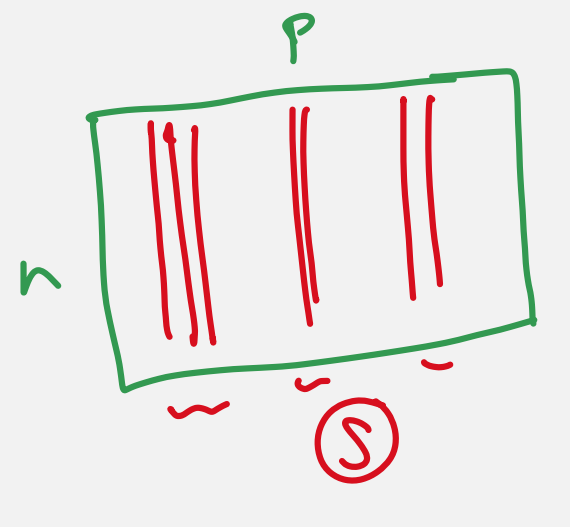
☺ When X is RIP for some (s, δ).

Restricted Isometry Property. (RIP).
Fix $\delta \in (0,1)$. A matrix $X \in \mathbb{R}^{n \times p}$ is (δ, s)-RIP if for every s-sparse β, it is the case that

$$(1-\delta)\|\beta\|_2^2 \leq \|X\beta\|_2^2 \leq (1+\delta)\|\beta\|_2^2.$$

Equivalent Cond X_S: $n \times s$ submatrix with columns indexed by S.

$$\max_{\substack{S \subset [p] \\ |S| = s}} \|X_S^T X_S - I\| \leq \delta$$



Today If $\boxed{X \text{ is RIP}} \Rightarrow$ a linear program recovers β^*.
LP

Thursday If $n = \Omega(s \log \frac{p}{s})$ and X has iid $\mathcal{N}(0,1)$ entries, then X is RIP.

High-Level Idea LP works if X is RIP with a small δ.

Thm (Candés–Romberg–Tao '04, Donoho–Tanner '05).
Suppose X satisfies RIP with order $2s$ and parameter $\delta = \delta_{2s} \in (0, \frac{1}{3})$. Then, for any s-sparse β^*, the unique solution to

no randomness in this statement

$$\text{(LP)} \quad \min_{z \in \mathbb{R}^p} \|z\|_1, \quad Xz = \underbrace{X\beta^*}_{Y}$$

ℓ_1-recovery.

is β^*.

(This LP finds the solution $Xz = Y$ with smallest ℓ_1-norm).

Remark LP is a convex program, so it can be solved "easily" (e.g., by LASSO).

Proof:

Lemma I. For every s-sparse u, v with disjoint supports,

$$|\langle Xu, Xv \rangle| \leq \delta_{2s} \cdot \|u\| \cdot \|v\|$$

if X is $(\delta_{2s}, 2s)$-RIP.

Pf $\langle u, v \rangle_{\mathbb{R}^p} = 0$, since u, v have disjoint supports.

$$|\langle Xu, Xv \rangle| = |\langle Xu, Xv \rangle - \langle u, v \rangle|$$
$$= |\langle X^T X u, v \rangle - \langle u, v \rangle|$$
$$= |\langle (X^T X - I) u, v \rangle_{\mathbb{R}^p}|.$$

Let S be the union of support of u and v. $|S| \leq 2s$

$$= |\langle (X_S^T X_S - I) u_S, v_S \rangle_{\mathbb{R}^{2s}}| \underset{\text{Cauchy Schwarz}}{\leq} \|(X_S^T X_S - I) u_S\| \cdot \|v_S\|$$

$$\leq \underbrace{\|X_S^T X_S - I\|}_{\leq \delta_{2s}} \cdot \underset{\|u\|}{\|u_S\|} \cdot \underset{\|v\|}{\|v_S\|}$$

(since X is $(\delta_{2s}, 2s)$-RIP).

So, $|\langle Xu, Xv \rangle| \leq \delta_{2s} \cdot \|u\| \cdot \|v\|$ ☒.

Lemma II (Nullspace Property).
Suppose for any $0 \neq v$ with $Xv = 0$ and any $S \subset [p]$ with $|S| = s$, $\|v_S\|_1 < \frac{1}{2}\|v\|_1$. Then, for any s-sparse β^*, the unique sol to LP is β^*.

ⓅⒻ Fix β that is S-sparse supported on $S \subset [p]$, $|S| = s$. $\{1, \ldots, p\}$.
Note that Take any $z \neq \beta$ with $Xz = X\beta$ $(= Y)$. By assumption, $z - \beta \neq 0$ and $X(z - \beta) = 0$.

$\|(z-\beta)_S\|_1 < \frac{1}{2}\|z - \beta\|_1$. Let $S^C = \{1, \ldots, p\} \setminus S$

$$\|(z-\beta)_S\|_1 < \frac{1}{2}\|(z-\beta)_S\|_1 + \frac{1}{2}\|(z-\beta)_{S^c}\|_1$$
$$= \frac{1}{2}\|z_S - \beta\|_1 + \frac{1}{2}\|z_{S^c}\| \rightarrow \beta = 0 \text{ outside } S.$$

$$\Rightarrow \|(z_S - \beta)\|_1 < \|z_{S^c}\|.$$

Now, triangle ineq

$$\|\beta\|_1 \leq \|\beta - z_S\|_1 + \|z_S\|_1$$
$$< \|z_{S^c}\|_1 + \|z_S\|_1 = \|z\|_1.$$

So, for any z with $Xz = X\beta$, $\|z\|_1 > \|\beta\|_1$.

Using Lemma II, it suffices to show that for any z with $Xz = 0$, $z \neq 0$ and any $S \subset [p]$, $|S| = s$,

$$\|z_S\|_1 < \frac{1}{2}\|z\|_1.$$

Partitioning of z: $z = (z_1, \ldots, z_p) \in \mathbb{R}^p$, order the coordinates of z: $|z_{i_1}| \geq |z_{i_2}| \geq |z_{i_3}| \geq \cdots \geq |z_{i_p}|$. Define

$S_0 = \{i_1, \ldots, i_s\}$, $S_1 = \{i_{s+1}, \ldots, i_{2s}\}$, $S_2 = \{i_{2s+1}, \ldots, i_{3s}\}$, ... etc.

It suffices to prove this for $\|z_{S_0}\| < \frac{1}{2}\|z\|_1$.

Write $z = z_{S_0} + z_{S_1} + \cdots$, z_{S_i} have disjoint supports.

$Xz = 0 \Rightarrow Xz_{S_0} = -X(z_{S_1} + z_{S_2} + \cdots)$ Moreover.

$2s$ order RIP $\Rightarrow$ s order RIP. So,

$$\|z_{S_0}\|_2^2 \leq \frac{1}{1-\delta_{2s}} \|Xz_{S_0}\|_2^2 \quad \text{(RIP)}.$$
$$= \frac{1}{1-\delta_{2s}} \langle Xz_{S_0}, Xz_{S_0} \rangle$$
$$= \frac{1}{1-\delta_{2s}} \langle Xz_{S_0}, -Xz_{S_1} - Xz_{S_2} - \cdots \rangle.$$

Notice $z_{S_0}, z_{S_1}, \ldots$ have disjoint supports. Thus, by Lemma-I

$$|\langle Xz_{S_0}, -Xz_{S_k} \rangle| \leq \delta_{2s} \|z_{S_0}\|_2 \|z_{S_k}\|_2, \quad \forall k \geq 1.$$

$$\|z_{S_0}\|_2^2 \leq \frac{1}{1-\delta_{2s}} \sum_{k \geq 1} |\langle Xz_{S_0}, -Xz_{S_k} \rangle|$$
$$\leq \frac{\delta_{2s}}{1-\delta_{2s}} \|z_{S_0}\|_2 \sum_{k \geq 1} \|z_{S_k}\|_2$$

Since $z \neq 0 \Rightarrow \|z_{S_0}\|_2 > 0$. Dividing by $\|z_{S_0}\|_2$,

$$\boxed{\|z_{S_0}\|_2 \leq \frac{\delta_{2s}}{1-\delta_{2s}} \sum_{k \geq 1} \|z_{S_k}\|_2.}$$

Claim. $\|z_{S_k}\|_2 \leq \frac{1}{\sqrt{s}} \|z_{S_{k-1}}\|_1$

ⓅⒻ Write $z_{S_{k-1}} = (a_1, \ldots, a_s)$, $z_{S_k} = (b_1, \ldots, b_s)$ with $|a_1| \geq \cdots \geq |a_s| \geq |b_1| \geq \cdots \geq |b_s|$.

Then, $\|z_{S_{k-1}}\|_1 = \sum_{i=1}^{s} |a_i| = \sqrt{\sum_{i,j} |a_i| \cdot |a_j|} \geq \sqrt{s \sum_i |b_i|^2}$
$\sqrt{s} \cdot \|z_{S_k}\|_2$.

$|a_i||a_j| \geq |b_\ell|^2$ for all $1 \leq i \leq j \leq s$
$1 \leq \ell \leq s$. ☒.

$$\|z_{S_0}\|_2 \leq \frac{\delta_{2s}}{1-\delta_{2s}} \frac{1}{\sqrt{s}} \sum_{k \geq 0} \|z_{S_k}\|_1 < \frac{1}{2\sqrt{s}} \underbrace{\sum_{k \geq 0} \|z_{S_k}\|_1}_{\|z\|_1}.$$

$$\delta_{2s} < \frac{1}{3} \Rightarrow \frac{\delta_{2s}}{1-\delta_{2s}} < \frac{1}{2}$$

Hence,
$\|z_{S_0}\|_2 \cdot \sqrt{s} < \frac{1}{2}\|z\|_1$ ★

Finally
$$\|z_{S_0}\|_2 \cdot \sqrt{s} = \sqrt{s(z_{i_1}^2 + \cdots + z_{i_p}^2)} \underset{\text{Cauchy Schwarz}}{\geq} \sqrt{(|z_{i_1}| + \cdots + |z_{i_p}|)^2} = \|z_{S_0}\|_1.$$

Hence. $\|z_{S_0}\|_1 < \frac{1}{2}\|z\|_1$

So, the null space property holds. $\Rightarrow$ Proves the result.